Slovenská Technická Univerzita – Fakulta elektrotechniky a informatiky, Ilkovičova 3, 812 19 Bratislava

Celulárny paralelný genetický algoritmus

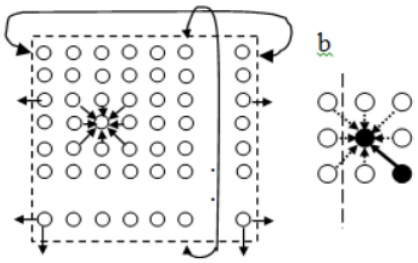
Umelá inteligencia 2, zadanie č. 3

Autor: **Matúš Nebus**
Dátum vypracovania: **6.10.2026**
Ročník štúdia: **1. Ing.**
Študijný program: **Robotika a kybernetika**

Nastavenia:

```
% SGA
sgaSize = 450;
bestNumsSGA = ones(1,18);
RandOldCountSGA = 90;
work1CountSGA = 162;
work2CountSGA = 180;
```

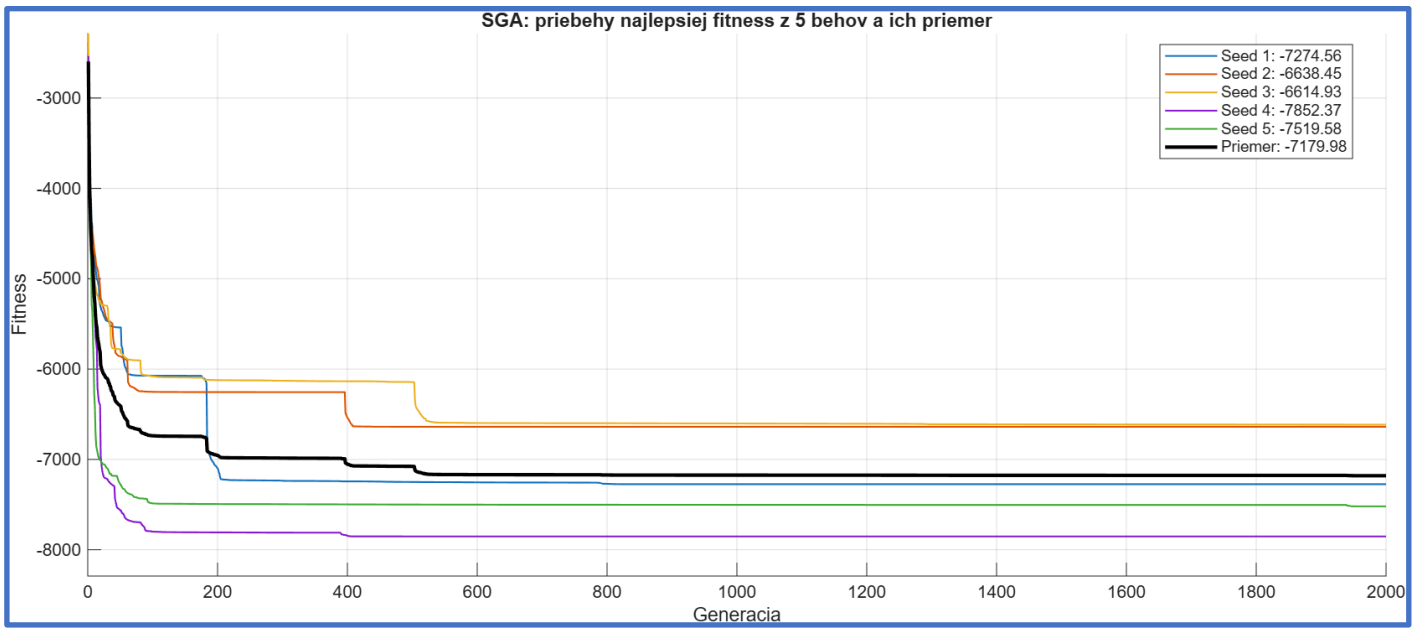
```
% PGA
numIslands = 9;
islandSize = 50;
migrationPeriod = 100;
bestNumsPGA = [1,1];
RandOldCountPGA = 10;
work1CountPGA = 18;
work2CountPGA = 20;
```

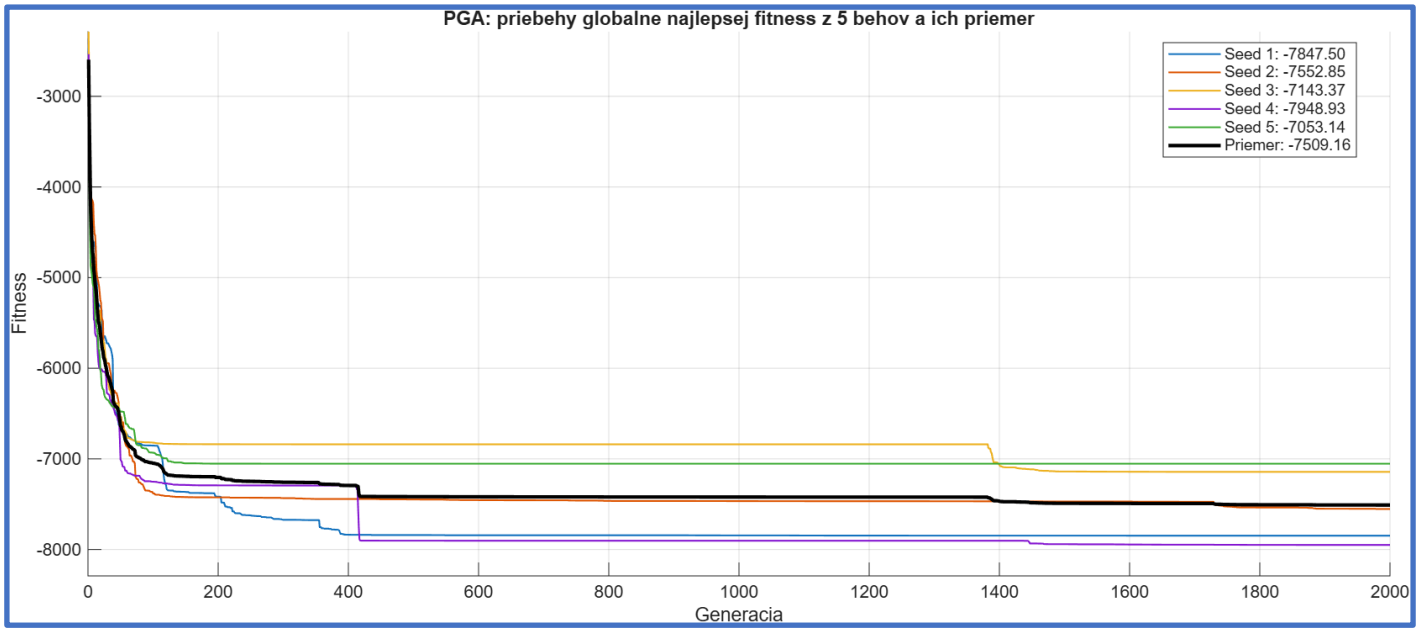


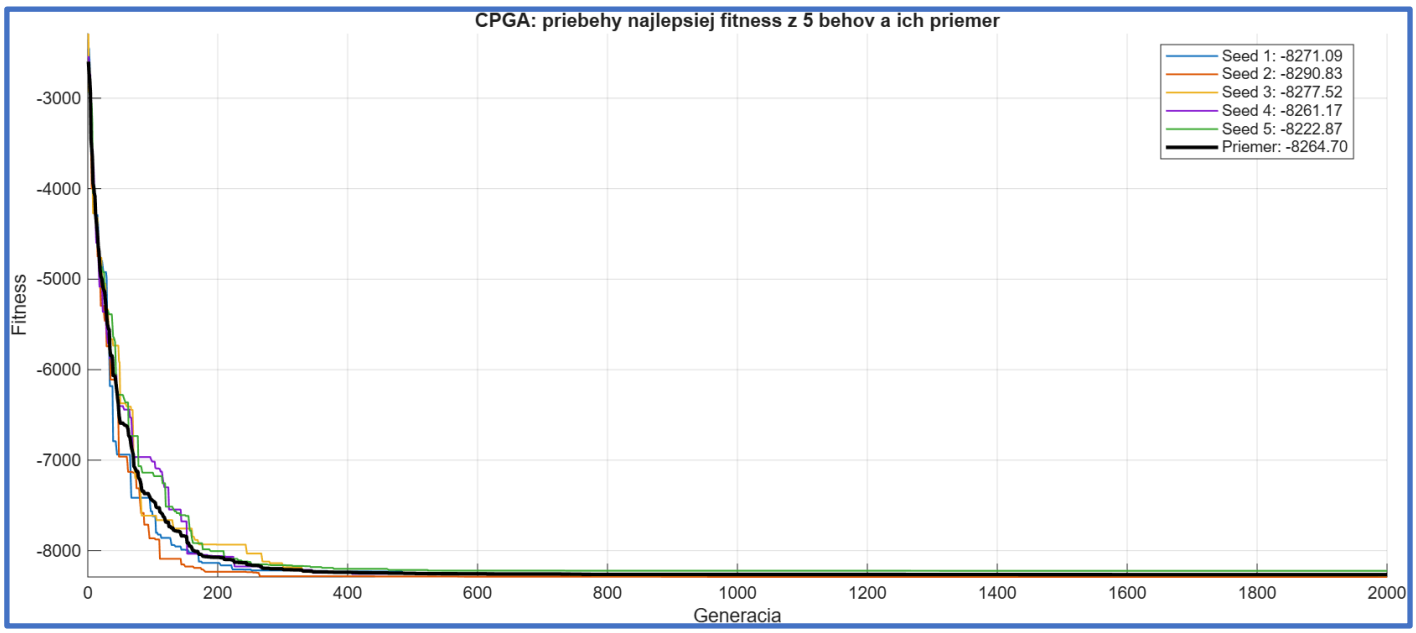

```
% CPGA
nRows = 30;
nCols = 15;
cpgaSize = nRows*nCols;
```

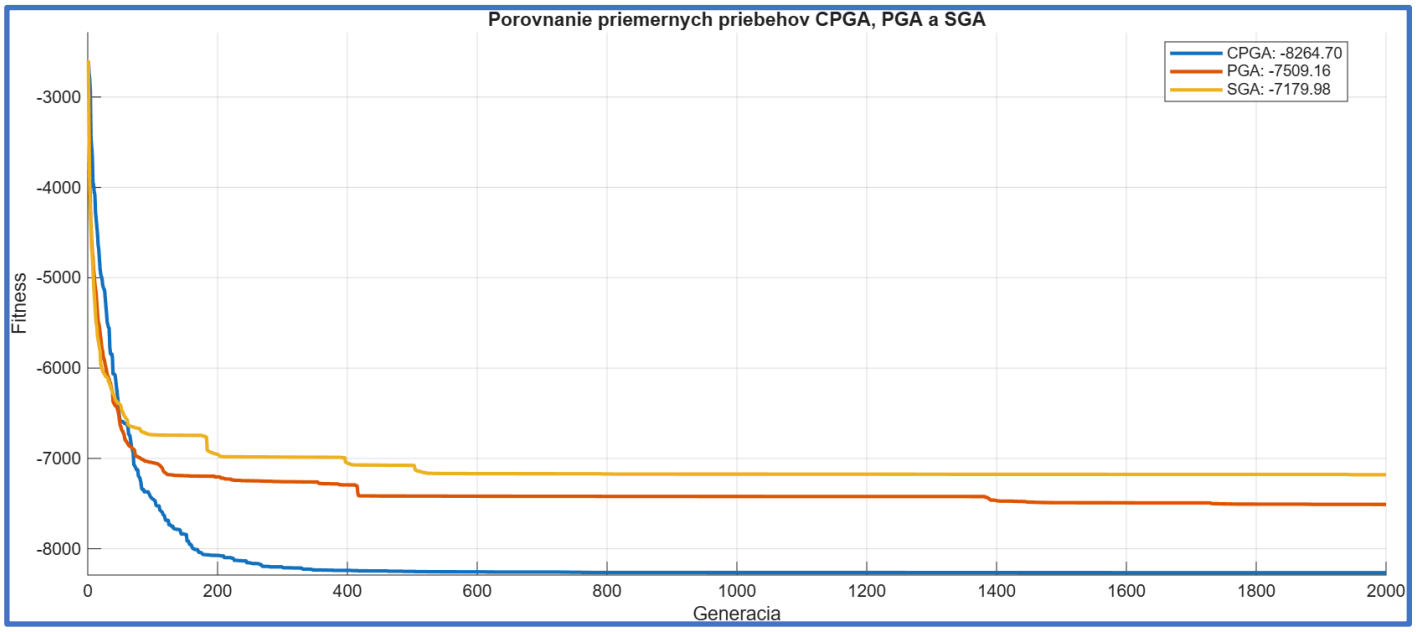
```
ngen = 2000;
pmut_global = 0.3;
pmut_local = 0.3;
Amp_local = 5 * ones(1,nvars);
```

Výsledky:









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TABULKA VYSLEDKOV

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Beh	CPGA	PGA	SGA
"Seed 1"	-8271.1	-7847.5	-7274.6
"Seed 2"	-8290.8	-7552.9	-6638.5
"Seed 3"	-8277.5	-7143.4	-6614.9
"Seed 4"	-8261.2	-7948.9	-7852.4
"Seed 5"	-8222.9	-7053.1	-7519.6
"Priemer"	-8264.7	-7509.2	-7180
"Smerodajna odchylka"	25.742	403.59	545.19

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NAJLEPSIE NAJDENE VEKTORY
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Algoritmus	Seed	Fitness	x1	x2	x3	x4	x5	x6	x7	x8	x9	x10
"CPGA"	2	-8290.8	481.18	431.75	445.17	457.56	472.14	427.63	442.31	455.08	469.64	425.17
"PGA"	4	-7948.9	-456.25	367.85	500	450.47	466.6	422.15	437.26	449.87	462.8	418.35
"SGA"	4	-7852.4	483.64	439.11	454.49	471.01	426.62	443.06	458.01	-298.4	-464.75	385.52